



\$\$\$ \textsf{\huge National Autonomous University of Mexico} \\\ \$\$\$

\$\$\$ \textsf{\huge Faculty of Engineering} \$\$\$

Hola!

This notebook is part of a series of geoscience applications designed to demonstrate practical approaches for solving problems using computational methods. It serves as a hands-on guide for understanding key concepts in energy-related subsurface systems through applied problem-solving rather than purely theoretical development. The focus is on building intuition and technical skills for modeling and analysis in real-world contexts, particularly within subsurface energy systems. These include petroleum reservoirs, geothermal resources, carbon storage and sequestration, and mineral exploration systems. By integrating fundamental principles with computational tools, the series aims to support learners in bridging the gap between geoscience theory and engineering practice, enabling more effective analysis and decision-making in complex subsurface environments.

Authors:

Dr. Moisés Velasco-Lozano

\$\$\$ \textsf{\huge Equilibrium constants \$K_i\$ of real gases} \$\$\$

1. Introduction

The equilibrium constants K_i are thermodynamic parameters that describe how a component is distributed between two phases at equilibrium, generally a liquid phase and a vapor phase. In a multicomponent system, the equilibrium constants K_i for a given component are defined as the ratio between the moles in the gas fraction y_i and the moles in the liquid fraction x_i . Mathematically, this relationship is expressed as:

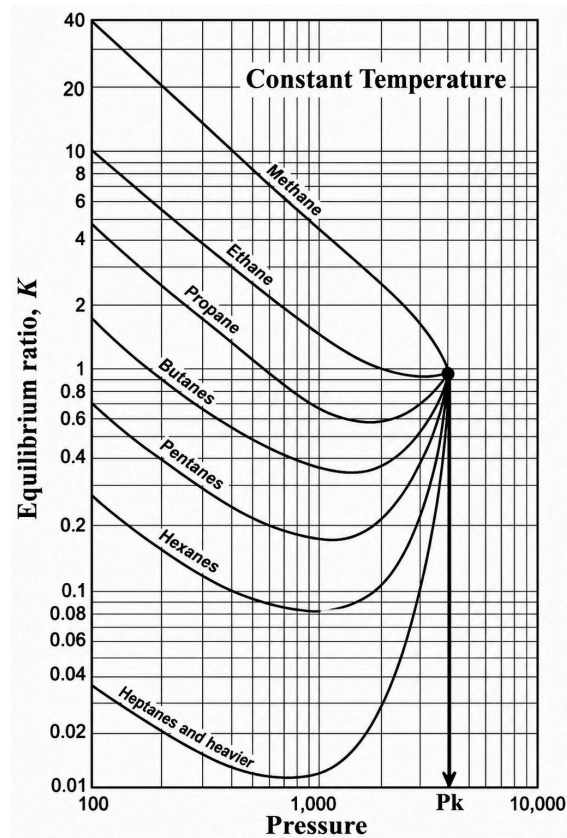
$$K_i = \frac{y_i}{x_i}$$

- These constants allow us to determine the tendency of each substance to evaporate or remain in the liquid phase. A high value of K_i indicates that the component is more volatile and preferentially concentrates in the vapor phase, whereas a low value indicates a greater affinity for the liquid phase.
- The concept of K_i is fundamental in the study of vapor–liquid equilibrium and in separation operations such as distillation, flash calculations, and absorption, since it facilitates the calculation of compositions and the design of industrial processes.

The values of equilibrium constants are reported in specialized charts for each pure component, which are functions of pressure and temperature. Likewise, there are correlations to estimate these values only at low pressures. A more rigorous approach is used to estimate equilibrium constants at high pressures and temperatures, where the concept of fugacity is included and complex iterative mathematical procedures are required. For a real solution, equilibrium constants are no longer solely a function of pressure and temperature, but also depend on the composition of the hydrocarbon mixture. This observation can be expressed by the following relationship:

$$K_i = K_i(p, T, z_i)$$

The behavior of the equilibrium constants is shown below.



Characteristics:

- The curves show a linear portion at low pressures.
- The curves never intersect.
- There is a trend with respect to the molecular weight of the components.

All curves converge at a pressure depending on the temperature under study; this pressure is called the convergence pressure, P_k

Numerous methods have been proposed for predicting equilibrium relationships in hydrocarbon mixtures. These correlations range from simple mathematical expressions to formulations involving a large number of composition-dependent variables. In this course, we will study some methods that provide adequate results when the mixture composition is not complex and the C7+ fraction is well characterized.

There are several numerical correlations available to compute the equilibrium constants, K_i , some of the most widely used include:

- Wilson
- Standing
- Whitson & Torp

2. Numerical solution

$$K_i = \left(\frac{p_{ci}}{p_k} \right)^{A-1} \left(\frac{p_{ci}}{p} \right) \exp \left[5.37A(1+\omega_i) \left(1 - \frac{T_{ci}}{T} \right) \right]$$


```
pc[9] = 436.9
pc[10] = 241
```

2.2.4 Critical temperature

```
In [138... # Temperature in Farenheit
Tc[0] = 87.82 #CO2
Tc[1] = -252.53 #N2
Tc[2] = -166.66
Tc[3] = 89.92
Tc[4] = 205.92
Tc[5] = 274.41
Tc[6] = 305.55
Tc[7] = 369
Tc[8] = 385.8
Tc[9] = 453.8
Tc[10] = 1330.19

# Temperature in Rankine
Tc = Tc + 460
```

2.2.5 Acentric factor

```
In [139... om[0] = 0.22369
om[1] = 0.0372
om[2] = 0.0104
om[3] = 0.0979
om[4] = 0.1522
om[5] = 0.1852
om[6] = 0.1995
om[7] = 0.228
om[8] = 0.2514
om[9] = 0.299
om[10] = 0.4419
```

2.2.6 Target temperature

```
In [140... # Temperature in Farenheit
T = 100

# Temperature in Rankine
T = T+460
```

2.2.7 Convergence pressure

```
In [141... a1 = 6124.3049
a2 = 2753.2538
a3 = 415.42049

pk = -2381.8542 + 46.341487 * MC7p * gC7p + a1*(MC7p*gC7p/(T-460))**1 + a2*(MC7p*gC7p/(T-460))**2 + a3*(MC7p*gC7p/(T-460))**3

print(pk)

14639.80849607037
```

2.2.8 Pressure range

```
In [134... p = np.linspace(100,pk,200) #psia

14639.80849607037
```

```
In [ ]:
```

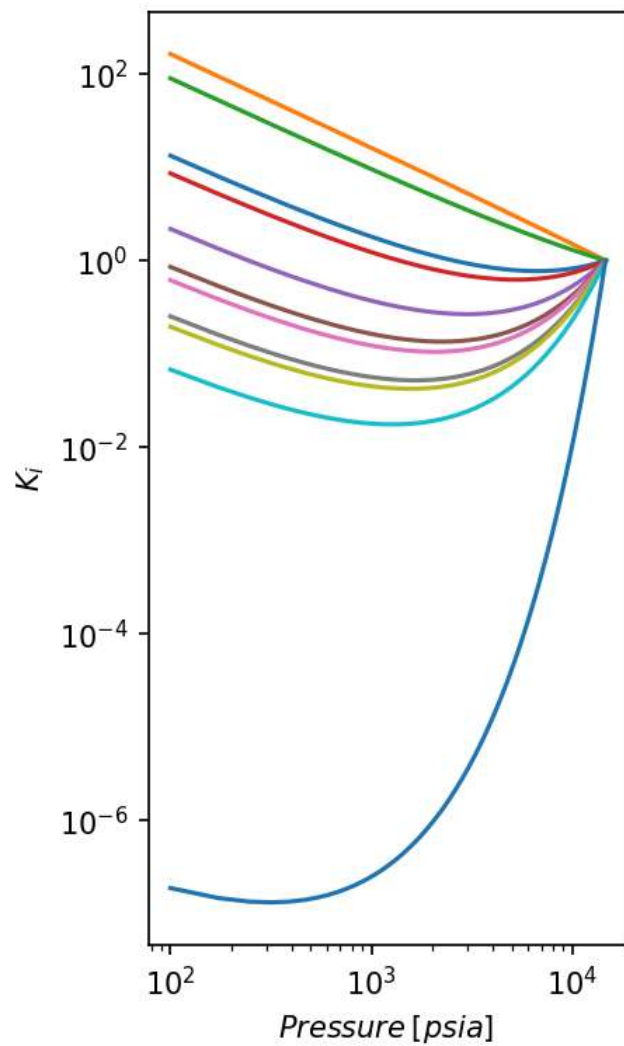
In [135... A = $1 - (p/p_k)^{0.7}$

2.3 Plotting

```
In [137... Ki=np.zeros(len(p))

plt.figure(dpi=150, figsize=(3,5))
for j in range(N):
    for i in range(len(p)):
        Ki[i] = (pc[j]/pk)**(A[i]-1)*(pc[j]/p[i])*np.exp(5.37*A[i]*(1+om[j])*(1-Tc[j]/T))
        # pre[:,j]=p[i]
    plt.loglog(p,Ki)
    plt.tight_layout()

plt.xlabel("$Pressure\":[psia]$ ")
plt.ylabel("$K_i$")
plt.show()
```



3. Conclusions and highlights

In []:

4. Comments

I hope this notebook is useful in your journey of coding applied to petroleum engineering and geosciences. I am a researcher interested in Machine learning for subsurface energy systems including oil & gas, carbon capture and storage, geothermal, and minerals. Please reach out!

Dr. Moisés Velasco-Lozano

Associate Professor

Petroleum Engineering Department

National Autonomous University of Mexico (UNAM)

mvelasco@ingenieria.unam.edu

mvelasco@unam.mx

In []: